FR-07 – Step 2: Identify equivalence classes for each input & output

Input: the variables and decisions (D1–D9) in `FR-07_1_variables.md`. Method: lecture S04 *Domain Testing*, step 2.

Guidelines applied (S04, slides 8–11)

Condition kind	Rule	Example in this feature
Range	1 valid class + 2 invalid classes (below, above)	Quantity 1 … M (M = maximum, probed at execution per D2)
Set	1 valid class per element (if handled differently) + 1 invalid class	Where "Add to cart" is clicked, dialog answer, reload type
Must be	1 valid class + 1 invalid class	Quantity is a whole number, product ID exists, token valid
Split	Split a class when its members may not be handled the same way	Quantity 0 vs negative, price 0/negative vs other wrong price, name with HTML

Legend: V = valid, I = invalid, TBD = expected result decided at execution. "Tester decision" = the spec is silent; the expected result was chosen by the tester (see the decision log). **M** = the maximum quantity. The spec has none; per D2 it is found at execution by probing from `99999999999` upward. Representative values are **not** chosen here; that is step 3.

Input equivalence classes

FR-07_IN_01 – Product being added

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_01	C1 Exists in catalog (must be)	Product exists in the catalog	V	Spec
FR-07_EC_02	C1 (split)	Product was deleted from the catalog (by admin) after being added to the cart	I	Tester decision: row removed or marked unavailable

In the UI a user can only add products that are listed, so "non-existent product" is tested through the API (IN_16).

FR-07_IN_02 – Quantity on the product detail page

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_03	C1 Required (must be)	Not empty	V	Assumption
FR-07_EC_04	C1 Required (must be)	Empty	I	Assumption
FR-07_EC_05	C2 Whole number (must be)	Plain whole number (digits only)	V	Spec (FR-06) + D9

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_06	C2 (split)	Decimal (e.g. `1.5`)	I	D9
FR-07_EC_07	C2 (split)	Non-numeric text (letters, symbols)	I	Spec (FR-06)
FR-07_EC_08	C2 (split)	Leading zeros (e.g. `007`)	V	Tester decision: accepted as 7
FR-07_EC_09	C2 (split)	With a plus sign (e.g. `+3`)	I	Tester decision: rejected
FR-07_EC_10	C2 (split)	Scientific notation (e.g. `1e2`)	I	Tester decision: rejected
FR-07_EC_11	C3 Range 1 … M	1 ≤ quantity ≤ M	V	Spec (FR-06: min 1) + D2
FR-07_EC_12	C3 Range, below (split)	Quantity = 0	I	Spec (FR-06)
FR-07_EC_13	C3 Range, below (split)	Quantity < 0	I	Spec (FR-06)
FR-07_EC_14	C3 Range, above	Quantity > M	I	D2

"Below the range" is split into 0 and negative: a negative quantity could be handled differently, for example by subtracting from the cart.

FR-07_IN_03 – Where "Add to cart" is clicked

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_15	C1 Source (set)	Product detail page (quantity from IN_02)	V	Spec (FR-06)
FR-07_EC_16	C1 Source (set)	Home page (quantity 1)	V	Assumption

No invalid class: there are no other "Add to cart" buttons.

FR-07_IN_04 – Product already in the cart

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_17	C1 In cart (set)	Not yet in the cart	V	Spec
FR-07_EC_18	C1 In cart (set)	Already in the cart, added again from the same page	V	Spec
FR-07_EC_19	C1 In cart (split)	Already in the cart, added again from the other page (home ↔ detail)	TBD	D8

FR-07_IN_05 – "+" button

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_20	C1 Result ≤ M (range)	Clicked when quantity < M	V	Spec
FR-07_EC_21	C1 Result ≤ M (range)	Clicked when quantity = M	I	D2

FR-07_IN_06 – "−" button

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_22	C1 Result ≥ 1 (range)	Clicked when quantity ≥ 2	V	Spec
FR-07_EC_23	C1 Result ≥ 1 (range)	Clicked when quantity = 1 (button must be disabled)	I	D3

FR-07_IN_07 – Delete button

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_24	C1 Clicked (set)	Delete clicked on a row	V	Spec

FR-07_IN_08 – Answer to the delete confirmation

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_25	C1 Answer (set)	Confirm	V	Spec
FR-07_EC_26	C1 Answer (set)	Cancel	V	Assumption

FR-07_IN_09 – "Continue shopping" button

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_27	C1 Clicked (set)	Clicked	V	Spec

FR-07_IN_10 – Number of distinct products in the cart

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_28	C1 Count (split)	0 products (empty cart)	V	Spec
FR-07_EC_29	C1 Count (split)	Exactly 1 product	V	Assumption
FR-07_EC_30	C1 Count (split)	2 or more products	V	Assumption

The count cannot be negative, so there is no invalid class. It is split because an empty cart has its own display (FR-07) and totals over several rows are computed differently from a single row.

FR-07_IN_11 – Login state

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_31	C1 Login state (set)	Guest (must be able to use the cart)	V	D1
FR-07_EC_32	C1 Login state (set)	Logged in	V	Spec

FR-07_IN_12 – Reload / leaving and returning

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_33	C1 Navigation type (set)	Navigate inside the app and come back (e.g. "Continue shopping", then Cart)	V	Spec (implied: FR-07 sends users shopping and merges re-added products)
FR-07_EC_34	C1 Navigation type (set)	Full page reload (F5)	V	Tester decision (D5): kept for logged-in users; lost acceptable for guests
FR-07_EC_35	C1 Navigation type (set)	Log out, then log in again	V	Tester decision (D5): kept

FR-07_IN_13 – `price` sent in `POST /api/cart`

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_36	C1 Equals catalog price (must be)	Same as the catalog price	V	D7
FR-07_EC_37	C1 (split)	A different positive price	I	D7
FR-07_EC_38	C1 (split)	Zero or negative	I	D7
FR-07_EC_39	C1 (split)	Non-numeric or missing	I	D7

FR-07_IN_14 – `name` sent in `POST /api/cart`

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_40	C1 Equals catalog name (must be)	Same as the catalog name	V	D7
FR-07_EC_41	C1 (split)	Different plain text	I	D7
FR-07_EC_42	C1 (split)	Contains HTML or script	I	D7 + SEC-04

FR-07_IN_15 – `quantity` sent in `POST /api/cart`

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_43	C1 Range 1 … M	Whole number, 1 ≤ quantity ≤ M	V	Assumption (FR-06 rule)
FR-07_EC_44	C1 Range, below (split)	0	I	Assumption
FR-07_EC_45	C1 Range, below (split)	Negative	I	Assumption
FR-07_EC_46	C1 Range, above	> M	I	D2
FR-07_EC_47	C2 Whole number (split)	Decimal	I	D9
FR-07_EC_48	C2 Whole number (split)	Non-numeric or missing	I	Assumption

FR-07_IN_16 – `id` sent in `POST /api/cart`

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_49	C1 Existing product (must be)	Existing product ID	V	Assumption
FR-07_EC_50	C1 (split)	Numeric ID of no product	I	Assumption
FR-07_EC_51	C1 (split)	Non-numeric or missing	I	Assumption

FR-07_IN_17 – Auth token on cart API

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_52	C1 Valid token (must be)	Valid token	V	Spec (API 4)
FR-07_EC_53	C1 (split)	No token	I	Spec (API 4)
FR-07_EC_54	C1 (split)	Invalid or tampered token	I	Spec (API 4)

FR-07_IN_18 – Quantity typed directly in a cart row

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_55	C1 Range 1 … M	Plain whole number, 1 ≤ quantity ≤ M	V	D6
FR-07_EC_56	C2 Required (must be)	Empty	I	D6

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-07_EC_57	C1 Range, below (split)	0	I	D6
FR-07_EC_58	C1 Range, below (split)	Negative	I	D6
FR-07_EC_59	C1 Range, above	> M	I	D6 + D2
FR-07_EC_60	C3 Whole number (split)	Decimal	I	D6 + D9
FR-07_EC_61	C3 Whole number (split)	Non-numeric text	I	D6
FR-07_EC_62	C3 Whole number (split)	Leading zeros (e.g. `007`)	V	Tester decision: accepted (as EC_08)
FR-07_EC_88	C3 Whole number (split)	Plus sign or scientific notation (`+3`, `1e2`)	I	Tester decision: rejected (as EC_09, EC_10)

EC_88 was split from EC_62 after review: the tester accepted `007` but rejected `+3` and `1e2`, so one class could no longer hold both. It is numbered after the output classes to keep earlier IDs stable.

Output equivalence classes

Class ID	Output variable	Equivalence class	Triggered by	Source
FR-07_EC_63	OUT_01 Table columns	Exactly Sản phẩm, Đơn giá, Số lượng, Thành tiền, Thao tác	Any non-empty cart (EC_29, EC_30)	Spec
FR-07_EC_64	OUT_02 Unit price	Catalog price, `₫` with thousand separators	Any add; EC_37–EC_39 via API	Spec
FR-07_EC_65	OUT_03 Quantity shown	Sum of quantities added, adjusted by +/− or typing	EC_11, EC_18, EC_20, EC_22, EC_55	Spec
FR-07_EC_66	OUT_04 Line total	Unit price × quantity, `₫` format	Any non-empty cart	Spec + Assumption
FR-07_EC_67	OUT_05 Cart total	Sum of line totals, `₫` format	EC_29, EC_30	Spec + Assumption
FR-07_EC_68	OUT_06 Total label	Exactly "Tổng cộng"	Any non-empty cart	Spec
FR-07_EC_69	OUT_07 Rows per product	Exactly 1 row	EC_18 (EC_19 if merge is decided)	Spec
FR-07_EC_70	OUT_08 Delete dialog	Shown before deleting	EC_24	Spec
FR-07_EC_71	OUT_09 Row after delete	Removed	EC_24 + EC_25	Spec
FR-07_EC_72	OUT_09 Row after delete	Kept, unchanged	EC_24 + EC_26	Assumption
FR-07_EC_73	OUT_10 After "Continue shopping"	Home page shown	EC_27	Spec
FR-07_EC_74	OUT_11 Button label	Exactly "Tiếp tục mua sắm"	Cart page shown	Spec
FR-07_EC_75	OUT_12 Empty state	Illustration/icon + clear message	EC_28 (including after deleting the last row)	Spec

Class ID	Output variable	Equivalence class	Triggered by	Source
FR-07_EC_76	OUT_13 Add feedback	Toast or badge update	EC_15, EC_16	Spec
FR-07_EC_77	OUT_14 Badge	Equals total units in the cart	Any cart change	Spec + D4
FR-07_EC_78	OUT_15 Invalid quantity error	Message shown; cart unchanged	EC_04, 06, 07, 09, 10, 12, 13, 14, 21, 56–61, 88	Assumption
FR-07_EC_79	OUT_16 Cart after navigation	Kept	EC_33, EC_34 (logged in), EC_35	Spec (implied) + D5
FR-07_EC_80	OUT_16 Cart after navigation	Lost	EC_34 (guest; acceptable, not a bug)	D5
FR-07_EC_81	OUT_17 API without valid token	Rejected (401/403); cart unchanged	EC_53, EC_54	Spec (API 4)
FR-07_EC_82	OUT_18 API with client price/name	Catalog values used; client values ignored	EC_37–EC_39, EC_41, EC_42	D7
FR-07_EC_83	OUT_18 API with invalid quantity/id	Request rejected; cart unchanged	EC_44–EC_48, EC_50, EC_51	Assumption
FR-07_EC_84	OUT_19 Delete button colour	Red	Cart page shown	Spec (FR-21)
FR-07_EC_85	OUT_20 Breadcrumb	Present on the Cart page	Cart page shown	Spec (FR-23)
FR-07_EC_86	OUT_21 Page heading	Exactly one `<h1>`	Cart page shown (empty and non-empty)	Spec (FR-21)
FR-07_EC_87	OUT_22 "−" at quantity 1	Disabled; row not removed	EC_23	D3

Summary

	Count
Valid input classes	30
Invalid input classes	32
TBD input classes (decided at execution)	1 (EC_19)
Output classes	25
Total	**88**

Decision log (spec silent, decided by the tester)

Bugs raised from these classes must say they rest on a tester decision, not on the spec.

Class	Question	Decision	Expected output
EC_02	A product in the cart is deleted from the catalog	**Invalid**	Row removed or marked unavailable; cannot be checked out
EC_08	Quantity `007`	**Valid**, accepted as 7	Quantity 7 added
EC_09	Quantity `+3`	**Invalid**	EC_78 (invalid quantity error)
EC_10	Quantity `1e2`	**Invalid**	EC_78

Class	Question	Decision	Expected output
EC_19	Same product from home page, then detail page (D8)	**Decided at execution**	Recorded with the result
EC_34	Full page reload (D5)	**Valid**: kept for logged-in users; lost is acceptable for guests	EC_79 (logged in) / EC_80 (guest)
EC_35	Log out and log in again (D5)	**Valid**: kept	EC_79
EC_62 / EC_88	Non-standard notation typed in the cart	Split: `007` **valid**; `+3`, `1e2` **invalid**	EC_65 / EC_78